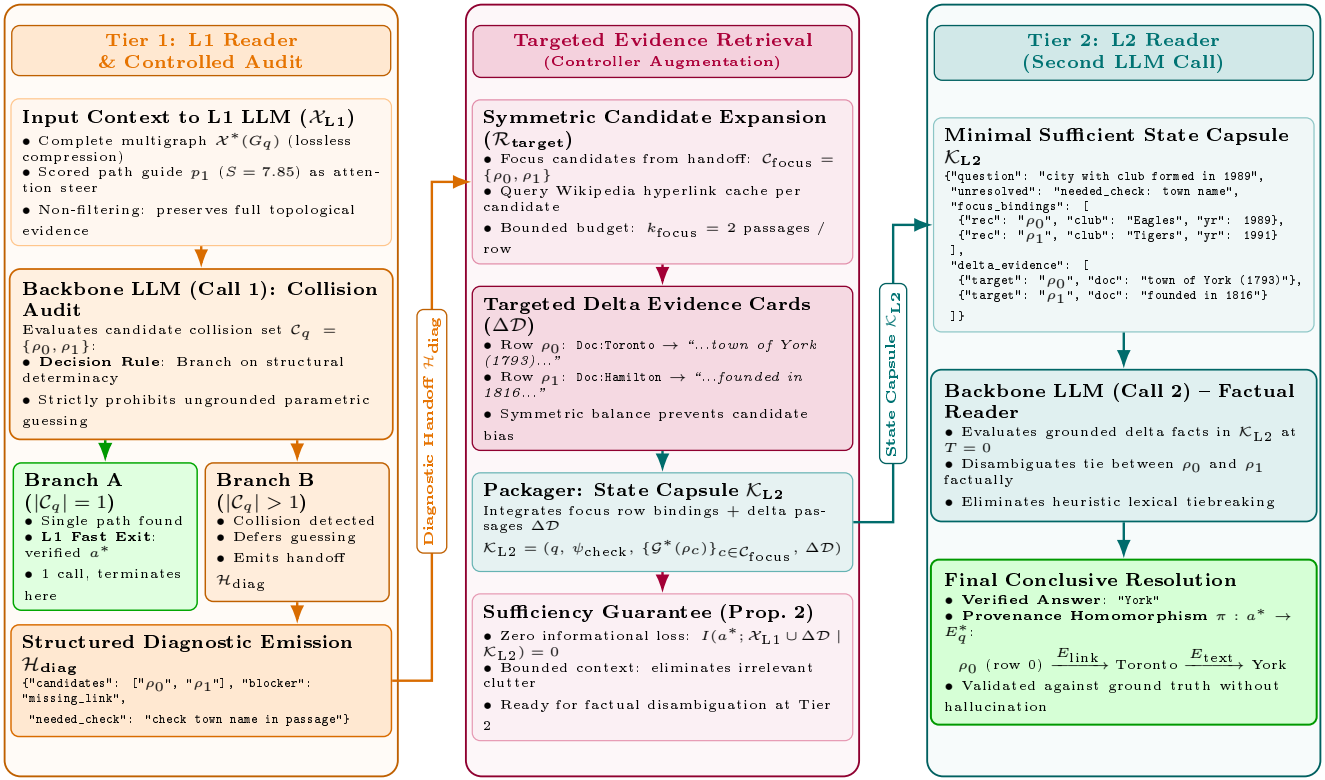

Tier 1: L1 Reader & Controlled Audit
Input Context to L1 LLM ($\mathcal{X}_{\mathbf{L1}}$)
• Complete multigraph $\mathcal{X}^*(G_q)$ (lossless compression)
• Scored path guide p_1 ($S = 7.85$) as attention steer
• Non-filtering: preserves full topological evidence
Backbone LLM (Call 1): Collision Audit
Evaluates candidate collision set $\mathcal{C}_q = \{\rho_0, \rho_1\}$:
• Decision Rule: Branch on structural determinacy
• Strictly prohibits ungrounded parametric guessing
Branch A ($|\mathcal{C}_q| = 1$)
• Single path found
• L1 Fast Exit: verified a^*
• 1 call, terminates here
Branch B ($|\mathcal{C}_q| > 1$)
• Collision detected
• Defers guessing
• Emits handoff $\mathcal{H}_{\text{diag}}$
Structured Diagnostic Emission $\mathcal{H}_{\text{diag}}$
{"candidates": ["ρ_0", "ρ_1"], "blocker": "missing_link",
"needed_check": "check town name in passage"}
Diagnostic Handoff $\mathcal{H}_{\text{diag}}$
Targeted Evidence Retrieval (Controller Augmentation)
Symmetric Candidate Expansion ($\mathcal{R}_{\text{target}}$)
• Focus candidates from handoff: $\mathcal{C}_{\text{focus}} = \{\rho_0, \rho_1\}$
• Query Wikipedia hyperlink cache per candidate
• Bounded budget: $k_{\text{focus}} = 2$ passages / row
Targeted Delta Evidence Cards ($\Delta\mathcal{D}$)
• Row ρ_0: Doc:Toronto → "...town of York (1793)..."
• Row ρ_1: Doc:Hamilton → "...founded in 1816..."
• Symmetric balance prevents candidate bias
Packager: State Capsule $\mathcal{K}_{\mathbf{L2}}$
Integrates focus row bindings + delta passages $\Delta\mathcal{D}$
$\mathcal{K}_{\text{L2}} = (q, \psi_{\text{check}}, \{\mathcal{G}^*(\rho_c)\}_{c \in \mathcal{C}_{\text{focus}}}, \Delta\mathcal{D})$
Sufficiency Guarantee (Prop. 2)
• Zero informational loss: $I(a^*; \mathcal{X}_{\text{L1}} \cup \Delta\mathcal{D} \mid \mathcal{K}_{\text{L2}}) = 0$
• Bounded context: eliminates irrelevant clutter
• Ready for factual disambiguation at Tier 2
State Capsule $\mathcal{K}_{\mathbf{L2}}$
Tier 2: L2 Reader (Second LLM Call)
Minimal Sufficient State Capsule $\mathcal{K}_{\mathbf{L2}}$
{"question": "city with club formed in 1989",
"unresolved": "needed_check: town name",
"focus_bindings": [
{"rec": "ρ_0", "club": "Eagles", "yr": 1989},
{"rec": "ρ_1", "club": "Tigers", "yr": 1991}
],
"delta_evidence": [
{"target": "ρ_0", "doc": "town of York (1793)"},
{"target": "ρ_1", "doc": "founded in 1816"}
]}
Backbone LLM (Call 2) – Factual Reader
• Evaluates grounded delta facts in $\mathcal{K}_{\text{L2}}$ at $T = 0$
• Disambiguates tie between ρ_0 and ρ_1 factually
• Eliminates heuristic lexical tiebreaking
Final Conclusive Resolution
• Verified Answer: "York"
• Provenance Homomorphism $\pi : a^* \to E_q^*$:
ρ_0 (row 0) $\xrightarrow{E_{\text{link}}}$ Toronto $\xrightarrow{E_{\text{text}}}$ York
• Validated against ground truth without hallucination